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ABSTRACT
This normative binding specifies the mapping of the respective eHealth DSI functional service onto the IHE Cross-Community Patient Discovery (XCPD) integration profile.

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1 Introduction

The proper identification of the patient at the point of care is a prerequisite for any data related transactions among countries in eHealth DSI. This normative binding specifies the mapping of the respective eHealth DSI functional service onto the IHE Cross-Community Patient Discovery (XCPD) integration profile.

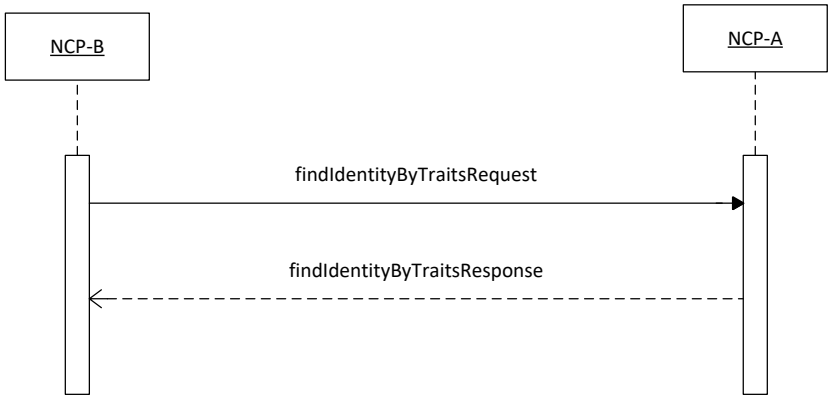
1.1 eHealth DSI Patient Identification and Authentication

Patient identification and authentication in eHealth DSI is considered to be end-to-end in a way that a health professional at a point of care in country-B makes use of patient management services within country-A, in order to obtain a valid patient identifier and to verify the authenticity of a patient. This interaction consists of two steps:

1. The patient provides the health professional at country-B with identity traits that are sufficient for authentication at the point of care and for identification within country-A.
2. The health profession makes use of eHealth DSI services to obtain a unique identifier for the patient. This identifier is shared between countries A and B and

MUST be used for any cross-country sharing of that patient's medical data which takes place within the context of the current encounter.

During the second step NCP-B transmit the provided identity traits to NCP-A and requests NCP-A for an identification of the patient and the provisioning of a patient identifier to be used by NCP-B in any further transactions that affect this patient's medical data, within the current encounter. The following figure shows this communication pattern as it is defined in [Interoperability Specifications] for the Patient Identification and Authentication Service.



The Patient Identification and Authentication Service is always provided by NCP-A and requested by NCP-B. The service contract consists of two messages: *FindIdentityByTraitsRequest* and *FindIdentityByTraitsResponse* where the response message is always triggered by the request message.

1.2 IHE Cross Community Patient Discovery (XCPD)

The eHealth DSI specification [Interoperability Specifications] defines the logical interfaces, the behaviour, and the overall semantics of the Patient Identification and Authentication Service but does not mandate for a specific technical standard for implementing the findIdentityByTraits request and response messages. This document specifies a respective binding onto the IHE Cross-Community Patient Discovery (XCPD) integration profile.

1.2.1 Profile Overview

In [IHE XCPD] the functionality and semantics of the XCPD integration profile is defined as:

The Cross-Community Patient Discovery (XCPD) profile supports the means to locate communities that hold patient relevant health data and the translation of patient identifiers across communities holding the same patient's data. A community is defined as a group of facilities/enterprises that have agreed to work together using a common set of policies for the purpose of sharing health information within the community via an established mechanism.

In the eHealth DSI context each participation nation takes the role of a community as described in the IHE XCPD scope definition. While community location is based on a static configuration in eHealth DSI, the IHE XCPD integration profile's functionality for translating and sharing patient identifiers is in the heart of the eHealth DSI Patient Identification and Authentication Service.

1.2.2 eHealth DSI Profiling of the XCPD Integration Profile

Within eHealth DSI the actors and transactions of the IHE XCPD integration profile are mapped onto eHealth DSI services and messages as follows:

IHE XCPD	eHealth DSI Patient Identification and Authentication Service
Actor: Initiating Gateway	NCP-B Patient Identification and Authentication Service consumer
Actor: Responding Gateway	NCP-A Patient Identification and Authentication Service provider
Transaction: Cross Gateway Patient Discovery ITI-55	FindIdentityByTraits request and response messages
Transaction: Patient Location Query ITI-56	*Not used by eHealth DSI*

This eHealth DSI binding introduces minor extensions and restrictions on the IHE XCPD actor and transaction definitions in order to properly cover the eHealth DSI use cases and to align with the eHealth DSI security framework:

- Additional error messages are defined that cover specific failure conditions of the eHealth DSI use cases
- The optionality of data fields is aligned to eHealth DSI privacy requirements

The application of security measures and the contents of the SOAP security header are specified normatively (see [Messaging Profile]).

1.3 Related Documents

The functional specification of the Patient Identification and Authentication Service is provided in [Interoperability Specifications] which is the normative foundation for all binding specifications. The technical binding defined in this document makes direct use of other bindings:

-
- [Audit Trail Profile] for a specification of the eHealth DSI common audit trail contents and structure
- [Messaging Profile] for a specification of how SOAP and Web Service Security are to be used for implementing eHealth DSI services
- [SAML Profile] for a specification of the HP Identity Assertion that is used for providing authentic information about the consumer of an eHealth DSI service

1.4 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].
eHealth DSI requirements are managed within a central requirements management tool and serialized into [eHealth DSI Requirements Catalogue]. References to requirements always use the identifiers as used with the central requirements. Newly defined requirements are handed over to the central requirements management. In case of doubt or conflict, [eHealth DSI Requirements Catalogue] MUST be considered to be the normative phrasing of a requirement.

1.5 Terms and Definitions

The country which holds information about a patient, where the patient can be univocally identified and where his data may be accessed is called "country-A" (country of affiliation) [eHealth DSI Glossary]. The country of treatment where cross-border health care is provided when the patient is seeking care abroad is called "country-B" (country of care) [eHealth DSI Glossary].
The term "Healthcare Professional (HP)" denotes a doctor of medicine, a nurse responsible for general care, a dental practitioner, a midwife or a pharmacist within the meaning of Directive 2005/36/EC, or another professional exercising activities in the healthcare sector which are restricted to a regulated profession as defined in Article

3(1)(a) of Directive 2005/36/EC, or a person considered to be a health professional according to the legislation of the country of treatment [eHealth DSI Glossary]. Health professionals are allowed to process medical patient data according to the legislation of the country of the health professional's residence. An "eHealth DSI Point of Care (PoC)" is a location where an eHealth DSI patient may seek healthcare services [eHealth DSI Glossary].
A "National Contact Point (NCP)" is an entity in each particular country to act as a bidirectional technical, organizational and legal interface between the existing national functions and infrastructures [eHealth DSI Glossary]. In this document the term "NCP" emphasizes the technical aspects of this interface and as such refers to a gateway which facilitates various aspects of cross-border data sharing (e.g. message forwarding, signature verification). From an architectural perspective NCPs denote the boundary between the eHealth DSI infrastructure and a country's existing national eHealth infrastructure (see [Interoperability Specifications] for details).

1.6 Namespaces

XML namespace prefixes are used in this document to stand for their respective namespaces as follows.

Prefix	Namespace
ehealth	urn:ehdsi:v1
soapenv	http://www.w3.org/2003/05/soap-envelope
wsse	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd
saml	urn:oasis:names:tc:SAML:1.0:assertion
hl7v2	urn:hl7-org:v2
hl7v3	urn:hl7-org:v3
xsi	http://www.w3.org/2001/XMLSchema-instance

1.7 Status of this Binding

The binding as defined in this document is a normative binding. All eHealth DSI countries MUST implement this binding within their NCP.
Other eHealth DSI documents must refer to this binding as [XCPD Profile]. Unless a version number is given, references to [XCPD Profile]. always refer to the most current version that is published on the eHealth DSI website.
Additional or alternative bindings MUST NOT be defined and implemented for the eHealth DSI.

2 eHealth DSI Patient Identification and Authentication Service

The *eHealth DSI Patient Identification and Authentication Service* is used to discover a valid patient identifier from an ID assigning authority by providing given identifiers and/or demographic data that is sufficient for patient identification.
The following specification of the *eHealth DSI Patient Identification and Authentication Service* is a profile on the IHE Cross-Community Patient Discovery integration profile [IHE XCPD].

2.1 findEntityByTraits Request Message

The *findIdentityByTraits()* request is initiated by an HP in the country of care for the identification of a foreign patient. The respective request message conforms to the *Patient Registry Find Candidates Query* (PRPA_IN201305UV02) interaction type as profiled by the IHE XCPD *Cross-Gateway Patient Discovery* transaction [IHE XCPD].

For the HL7 transmission wrapper and the *HL7 Control Act* the conventions identified in IHE ITI TF appendix O [IHE ITI TF-AppendixO] and the changes from the XCPD supplement [IHE XCPD] appendix O MUST be followed.
In addition the following eHealth DSI-specific restrictions apply:

- <receiver/> MUST refer to NCP-A. Other sub-elements than the device-identifier that holds the OID of NCP-A MUST be ignored by the service provider and SHOULD NOT be provided by the service consumer. <sender/> MUST refer to NCP-B. Other sub-elements than the device-identifier that holds the OID of NCP-B MUST be ignored by the service provider and SHOULD NOT be provided by the service consumer.
- Asynchronous operations MUST NOT be used. According to [Messaging Profile] all message interchange in eHealth DSI MUST be synchronous.
- Demographic Query Only mode or Shared/national Patient Identifier Query and Feed mode MUST be used. Other modes defined in [IHE XCPD] MUST NOT be used.
- The health data locator option as defined in section 27.2.1 of [IHE XCPD] MUST NOT be used. Where indication of support for the health data locator option is required in responses, the service provider MUST provide the value "NotHealthDataLocator".
- The revoke option as defined in section 27.2.2 of [IHE XCPD] MUST NOT be used.
- Correlations MUST NOT be cached by the service provider. The respective syntax elements described in section 3.55.4.1.2 of [IHE XCPD] MUST NOT be used.

Reverse Cross-Gateway Queries MUST NOT be used. The *homeCommunityId* and *community patient id assigning authority* arguments SHOULD be set to the OID of the responding NCP (NCP-A) in query requests.

2.1.1 Definition of National Identification Traits

Each NCP-A SHALL define its own configuration file (EHEALTH-107) specifying the traits required for patient identification using the eHDSI Central Services according the [Service Location and Capability Lookup Profile] and conformant with the eHDSI Insernational Search Mask schema.

2.1.2 Restrictions on the Use of Traits

For a *findIdentityByTraits()* request only the following traits or a subset of these MUST be used. Service providers SHOULD reject requests that contain other traits than the ones listed below. For detailed information on the encoding of these traits and their optionality for the XCPD modes see [IHE XCPD].

Identity Trait	Source	Usage Convention (if provided)
LivingSubjectID	Personal ID Card	SHOULD contain zero or more living subject Id. When present, it shall contain both an assigning authority identifier (root) and individual ID (extension). If multiple subject IDs are given for the same patient, each identifier MUST be provided as a dedicated <LivingSubjectID/> element (for instance National or Regional ID).
LivingSubjectName	Personal ID Card	Family name and given name MUST both be given if no LivingSubjectID is provided. Otherwise this query parameter is optional.
LivingSubjectBirthTime	Personal ID Card	Birth date MUST be given if no LivingSubjectID is provided. Otherwise this query parameter is optional. If given this parameter MUST be encoded as "YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][+/-ZZZZ]" with year, month and day being mandatory.

LivingSubjectAdministrativeGender		MUST be "M", "F" or "UN"
PatientAddress	Personal ID Card	SHOULD contain country and city.

2.1.3 Use of Pseudonyms and Temporal Identifiers

A country MAY wish to protect its patients' privacy by negotiating an eHealth DSI shared identifier from a pseudonymous or temporal national patient identifier. In this case *Shared/national Patient Identifier Query and Feed mode* MUST be used. On successful identification within country A the response message MUST at least provide the patient's date of birth in order to allow the HP in country B to verify the accuracy of the identification.

2.1.4 Patient Authentication

The *eHealth DSI Identification Service* allows for the identification of a patient. If a country requires an additional authentication of its citizens when they ask for medical care in another country, this country MUST define its own authentication service. The *eHealth DSI Identification Service* findIdentityByTraits operation only provides the mechanism for transparently transporting authentication data alongside with the transactions (piggybacking) between NCPs. This is done by using two HL7v3 *instance identifier* within a single <LivingSubjectID/> element; one identifier is used for identifying the patient while the other one is used for authentication of the patient.
The service provider MUST be able to distinguish identifier and authentication object by their roots (assigning authorities).

2.1.5 Request Accuracy of Matches

The *HL7 Patient Registry Find Candidates Query* allows to further refine the match criteria by setting the match algorithm and specifying a requested minimum degree of match for the provided traits.
As the respective <MatchCriterionList> element is optional with the HL7 schema, it SHOULD NOT be used for eHealth DSI. If present, the minimum requested match degree SHOULD be set to an integer value of "100". In both cases the responding service SHOULD only respond with identity data of patients who fully match all provided traits. Returning multiple candidates' identity trails SHOULD be avoided for privacy reasons.

2.1.6 Next of Kin

The patient's identification data are needed when searching for a patient. They are composed of a subset of the patient's demographic data. For member states implementing the authorized third party option, an authorized third party can act on behalf of the patient. In this situation, authorized third party identification data are also required, to allow the Country of Affiliation to verify the authorization.
The information of the patient (the Patient Identification Data) and the information of the third party, must both be communicated to the Country of Affiliation, at the same time.
Each NCP-A SHALL define its own configuration file (EHEALTH-107) specifying the traits required for the next of kin identification using the eHDSI Central Services according the [Service Location and Capability]and conformant with the eHDSI Insernational Search Mask schema.

2.2 *Information related to the Next of Kin are ensured through a dedicated SAML assertion defined in the [SAML Profile].* Expected Actions at NCP-A

The *eHealth DSI Identification Service* provider shall respond with the *findEntityByTraits* response message containing the patient identifier that is to be used for querying the identified patient's medical data. The *eHealth DSI Identification Service* provider MUST verify that the requesting service user has sufficient rights to query for the identifier of the given patient. It is subject to the national security policy of the patient's country of affiliation, how multiple matches and matches with less than 100% accuracy are handled.
In case of an error that relates to the transmission of the request or the processing of the eHealth DSI security token, the *eHealth DSI Identification Service* provider MUST respond with a fault message according to [Messaging Profile].

2.3 FindIdentityByTraits Response Message

The *eHealth DSI findEntityByTraits response* content is based on *HL7 Patient Registry Find Candidates Query Response (PRPA_IN201306UV02)* interaction, as profiled by the *IHE XCPD Cross-Gateway Patient Discovery* result message.
For the HL7 transmission wrapper and the *HL7 Control Act* the conventions identified in the IHE PIX/PDQV3 supplement appendix O and the changes from the XCPD supplement appendix O MUST be followed.
In addition the following eHealth DSI-specific restrictions apply:

- <receiver/> MUST refer to NCP-B. Other sub-elements than the device-identifier that holds the OID of NCP-B MUST be ignored by the service consumer and SHOULD NOT be provided by the service provider. <sender/> MUST refer to NCP-A. Other sub-elements than the device-identifier that holds the OID of NCP-A MUST be ignored by the service consumer and SHOULD NOT be provided by the service provider.
- Asynchronous operations MUST NOT be used. According to [Messaging Profile] all message interchange in eHealth DSI MUST be synchronous.
- Correlations MUST NOT be cached by the service provider. The respective syntax elements described in section 3.55.4.1.2 of [IHE XCPD] MUST NOT be used.

The <processingCode/> MUST be set to "D" (debugging) for PPT. It MUST be set to "P" (production) for eHealth DSI and routine operations.

2.3.1 Full Success Scenario

For each matching candidate a single <subject/> element MUST be included within the control act wrapper.
In addition to the constraints defined in [IHE XCPD] the following conventions MUST be followed for <subject1/patient/> elements:

Element Name	Opt.[1]	eHealth DSI Usage Convention
patient	R	For each matching candidate a single <patient/> element MUST be provided.
patient/id	R	This element MUST contain the HL7-II-encoded Id of the patient that MUST be used for subsequent transactions to access the patient's medical data. The root designator MUST be present.
patient/statusCode	R	MUST be "active".
patient/patientPerson	R	Additional demographic data on a patient that matches the query. The encoding of this data MUST follow the conventions as stated in [IHE XCPD].

[1] Categories MUST be filled (**R**), which MAY be filled (**O**) and which categories MUST NOT be used (**X**).

		See table below for a list of demographics that SHOULD be used for eHealth DSI.
patient/subjectOf1/ queryMatchObservation	R	This element encodes the score of the match as an HL7 observation. It MUST be used as this: <hl7v3:queryMatchObservation classCode="OBS" moodCode="EVN"> <hl7v3:code codeSystem="2.16.840.1.113883.1.11.19914"/> <hl7v3:value xsi:type="hl7v3:INT" value="***MATCH***"/> </hl7v3:queryMatchObservation>

Other elements MAY be provided within the result set by the sender but SHOULD be ignored by the receiver.
For a FindIdentityByTraits response only the following ID data MUST be provided as child elements of the <patientPerson /> element.

Identity Data	Opt.	Usage Convention (if provided)
asOtherIDs/id	O	This element SHOULD be only given if it provides further information on the scope and context of the used identification mechanism. This information SHOULD be suited to allow the HP to verify the claimed identity of the patient.
name	O	Both family name and given name SHOULD be provided. Note: This element is mandatory wrt. the HL7v3 schema. Therefore at least an empty instance MUST be included with the response.
birthTime	R+	MUST be provided.
birthplace	O	SHOULD contain the country and city of birth
administrativeGenderCode	O	
addr	O	
guardian	X	For the eHealth DSI pilots minors and dependent people will not be treated different from others. This element MUST NOT be provided as no respective risk assessment has been done.

As specified in [IHE XCPD], the following status should be returned:

- AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper).

OK (data found, no errors) is returned in QueryAck.queryResponseCode (control act wrapper).

2.3.2 eHealth DSI Extended Security Safeguards (ESS)

eHealth DSI ESS may in certain scenarios piggyback security related data with the Patient/id attribute. The "@@"-phrase is used as a delimiter between the patient id and the additional ESS related information.

```
<patient classCode="PAT">
  <id extension="13831630766093022953 78@@1234ABCD" root="2.16.17.710.790.1000.990.1"/>
  […]
</patient>
```

In order to allow for parallel execution of non-ESS and ESS-enabled NCPs the following constraints are defined on patient identifiers as provided by NCP-A:

- eHealth DSI implementations MUST NOT piggyback any further data with patient identifiers except the key index information as specified in [Security Services Specification]

If NCP-B receives a XCPD response message that includes a "@@"-phrase, an error message MUST be returned to NCP-A as this scenario indicates that country-A requests for ESS while NCP-B does not implement an ESS-Manager (which would have intercepted the response message and would have removed the key index from the message).

2.3.3 Error handling

If the *eHealth DSI Identification Service* provider does not find a matching patient identifier it SHOULD include a <reasonOf/> element with the response message:

```
<reasonOf typeCode="RSON">
  <detectedIssueEvent classCode="ALRT" moodCode="EVN">
    <code code="ActAdministrativeDetectedIssueCode" codeSystem="2.16.840.1.113883.5.4"/>
    <!-- details on detected issue and proposed activity -->
  </detectedIssueEvent>
</reasonOf>
```

Depending on the reason for not providing a patient identifier, the codes and messages as defined below MUST be used[2]:

Condition and proposed action	Reason Encoding
The service requestor tried an identification based on an ID only or did not provide enough data to univocally identify the patient. (WARNING). The HP SHOULD ask the patient for further demographics and re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). OK (data found, no errors) is returned in QueryAck.queryResponseCode (control act wrapper).	<triggerFor typeCode="TRIG"> <actOrderRequired classCode="ACT" moodCode="RQO"> <code code="AdditionalDemographicsRequested" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </actOrderRequired> </triggerFor> If specific demographics are requested the respective code values of code system 1.3.6.1.4.1.19376.1.2.27.1 as specified in section 3.55.4.2.2.6 of [IHE XCPD] SHOULD be used. There may be as many triggerFor elements, each of them containing an ActOrderRequired element as needed to code the attributes which would increase the assurance of the match[3].
The service provider only allows for patient identification by national/shared ID (WARNING). The HP SHOULD ask the patient for a national (health care) identification card and re-issue the request using *Shared/national Patient Identifier Query and Feed mode*. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<triggerFor typeCode="TRIG"> <actOrderRequired classCode="ACT" moodCode="RQO"> <code code="DemographicsQueryNotAllowed" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </actOrderRequired> </triggerFor>
The service provider only allows for patient identification by national health card or EHIC. Queries based on demographics only are not supported (WARNING) The HP SHOULD ask the patient for a health care identification card and re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<triggerFor typeCode="TRIG"> <actOrderRequired classCode="ACT" moodCode="RQO"> <code code="EHICDataRequested" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </actOrderRequired> </triggerFor>
The service provider does not accept the query because responding MAY lead to a disclosure of private patient data (ERROR). The HP SHOULD limit the provided traits and re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper).	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="EVN"> <code code="PrivacyViolation" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </detectedIssueManagement> </mitigatedBy>

[2] All codes using the coding system: codeSystem="1.3.6.1.4.1.19376.1.2.27.3 are to be used per XCPD error code definition.

[3] See IHE ITI CP #535

AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	
The requestor has insufficient rights to query for patient's identity data (ERROR). If access to the patient's medical data is required at the PoC this MUST be performed by a person with additional permissions. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="EVN"> <code code="InsufficientRights" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </detectedIssueManagement> </mitigatedBy>
Patient authentication MUST be piggybacked with patient identification. A respective identifier (e.g. GSS TAN) was not provided (ERROR) The HP at the PoC SHOULD ask the patient for a respective identifier and SHOULD re-issue the request. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="EVN"> <code code="PatientAuthenticationRequired" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </detectedIssueManagement> </mitigatedBy>
The service provider did not find a match with the given minimum accuracy. (INFO) The service consumer SHOULD re-issue the request with a lower minimum confidence level. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). OK (data found) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="EVN"> <code code="AnswerNotAvailable" codeSystem="1.3.6.1.4.1.19376.1.2.27.3"/> </detectedIssueManagement> </mitigatedBy>
The identity traits provided by the service consumer are not supported by the service provider. (ERROR) The service consumer SHOULD re-issue the request with a different set of identity traits. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="EVN"> <code code="AnswerNotAvailable" codeSystem="1.3.6.1.4.1.19376.1.2.27.3"/> </detectedIssueManagement> </mitigatedBy>
The service consumer defined a confidence level that conflicts with the security policy of the service provider. (INFO) The service provider SHOULD respond only with the candidate matches that it is allowed to provide wrt. its security policy. AA (application accept) is returned in Acknowledgement.typeCode (transmission wrapper). AE (application error) is returned in QueryAck.queryResponseCode (control act wrapper)	<mitigatedBy typeCode="MITGT"> <detectedIssueManagement classCode="ACT" moodCode="EVN"> <code code="PolicyViolation" codeSystem="1.3.6.1.4.1.12559.11.10.1.3.2.2.1"/> </detectedIssueManagement> </mitigatedBy>

Additional information to describe the reason MUST be included in the acknowledgementDetail of the response.

AcknowledgementDetail.Code	The value MUST be a code listed in the tables of the Appendix 6.4 of [NCPeH Components Specifications].https://webgate.ec.europa.eu/fpfis/wikis/x/n_PzN
AcknowledgementDetail.Text	The value is a text description of the error code
AcknowledgementDetail.Location	The location tag SHOULD be used to pass custom information from the NC-A to the NC-B.

Below a example of the tab acknowledgementDetail :

```
<acknowledgementDetail>
        <code code="ERROR_PI_NO_MATCH"/>
        <text> The identification and authentication used in the Country of Treatment (Country B) are not correctly provided or does not match any existing patient.^ additional custom information from country A
        </text>

<location>XCPDServiceImpl.parsePRPAIN201305UV02toPatientDemographics(100)</location>
 </acknowledgementDetail>
```

3 Security Considerations

3.1 Application of eHealth DSI Security Services

Implementations of this binding MUST consider the eHealth DSI standard security safeguards in order to preserve basic protection for data confidentiality, data integrity and patient privacy:

- All messages MUST make use of the Web Service Security Standard and the WS Addressing Standard as defined in [Messaging Profile].
- Each request message MUST contain claims on the identity of the requestor and on the care context as defined in [SAML Profile] and [Messaging Profile].
- The document recipient MUST verify the integrity and authenticity of any assertions and other security token communicated alongside for each incoming request.
- Before accepting provided data the document recipient MUST verify the existence of a valid consent and assess all applicable security policies.

3.2 XCPD Specific Security Measures

The national security policy of the patient's country of affiliation MUST always override weaker service consumer minimum confidence level: for instance, if country A only accept 100% match but country B is requesting with minimum confidence level of 75%, then only 100% matches will be returned.

3.3 Security Audit Considerations

Both the eHealth DSI Identification Service Provider and Consumer write an audit trail entry according to the ID Mapping audit schema as defined in [Audit Trail Profile].

eHealth DSI Instance	Opt.	Description
Event	R	Audited event
Human Requestor	R	HP who triggered the event
Source Gateway	R	Service consumer node address at the country of care
Target Gateway	R	Service provider node address at the country of affiliation
Mapping Service	R / X	Service that provided the mapping. MUST be filled by the service provider. MUST NOT be filled by the service consumer.

Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient Source	R	Patient whose identifier was discovered or mapped
Patient Target	R	Result of the mapping operation
Error Message	O	Only used in case that the transaction was not completed successfully or for anomaly detection

Table 1: eHealth DSI Patient Identification Service Audit Message Categories

4 References

4.1 Normative References

[IHE ITI TF-Appendix0] IHE International: IHE IT Infrastructure (ITI) Technical Framework. Appendix 0: HL7 v3 Transmission and Trigger Event Control Act Wrappers. https://profiles.ihe.net/ITI/TF/Volume2/ch-O.html

[IHE XCPD] IHE International: Cross-Community Patient Discovery. ITI TF Supplement for Trial Implementation. August 2012. http://www.ihe.net/Technical_Framework/upload/IHE_ITI_Suppl_XCPD_Rev2-4_TI_2012-08_31.pdf

[RFC 2119] Bradner, S.: Key words for use in RFCs to Indicate Requirement Levels. Harvard University, Boston, Massachusetts, 1997.

5 Appendix A: Example Messages

5.1 Appendix A.2: FindIdentityBy Traits Based on EHIC Data

The following example shows the mapping of Czech EHIC data elements onto a PRPA_TE201305UV02 control act.

